

Cover Letter

To: Professor Roman Słowiński, Co-ordinating Editor-in-Chief

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Re: New submission — *Online Traveling Salesman Games: Core Fragility and the Temporal Nucleolus under Dynamic Arrivals*

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Dear Professor Słowiński,

We submit the enclosed manuscript for consideration as a new research article in *EJOR*. The paper extends the cooperative Traveling Salesman Game of Potters, Curiel, and Tijs (1992) to an online setting in which customers arrive over time, couples the cooperative-game stability question to the realized dispatch dynamics, and provides both analytic and computational results.

The topic sits squarely at the intersection of routing and cooperative game theory, a combination with strong *EJOR* lineage: the basic Vehicle Routing Game nucleolus is from Göthe-Lundgren, Jörnsten, and Värbrand (1996, *Mathematical Programming*); the heterogeneous-fleet extension and the time-window VRG nucleolus appeared respectively in *Transportation Science* (Engvall et al. 2004) and *Applied Mathematical Modelling* (Tae, Kim, and Park 2020); the cooperative-TSP rolling-horizon Shapley/Core variants are from Kimms and Kozeletskyi (2016a, *EJOR*; 2016b, *EURO J. Transp. Logist.*); and the most recent collaborative-routing cost-allocation paper (Lyu, Lalla-Ruiz, and Schulte, 2025, *EJOR*) provides the most direct methodological precedent — row-generation over a static coalition family, replaced here by a sequential-LP cascade over a *time-induced* family.

Core contribution

We introduce the *Online Traveling Salesman Game* $(N, c, \mathcal{F})$, a cooperative game in which the family $\mathcal{F}$ of feasibility- admissible coalitions is endogenously induced by the realized arrival-service process rather than specified exogenously as in Myerson-style graph-restricted or antimatroid-restricted games. On this family we define the *Temporal Nucleolus* as the lexicographic minimizer of excess and compute it by a sequential linear program over $\mathcal{F}$.

The paper’s theoretical contribution is three complement-family sufficient conditions for Temporal Core emptiness — a single- complement threshold r^{**} (Theorem 9), a balanced-complement threshold r^{***} (Proposition 10), and a balanced-near-complement threshold $r^{(\diamond)}$ on the mixed collection $B_{n-1} \cup B_{n-2}$ (Proposition 13) — together with one unifying partition-pair certificate (Proposition 14) that contains Theorem 9 as the singleton- complement case and additionally fires on non-singleton partition pairs at $k < n - 1$. A structural obstruction (Corollary 18): a bounded peak queue $k < n - 1$ simultaneously blocks all three complement-family mechanisms. Together the four analytic thresholds certify 66 of the 67 observed empty Cores in our 175-instance grid (the remaining one near-complement case eludes the closed-form thresholds by $\approx 2 \times 10^{-3}$), with no false

positives across 525 policy-instance pairs (three dispatch policies). An asymptotic refinement (Theorem 16) ties $r^{**} \rightarrow 1$ to the classical Beardwood–Halton–Hammersley $\sqrt{n}$ scaling of the Euclidean TSP, and a scale-up to $n = 50$ confirms the asymptotic decay.

Scope and methodological choice

The paper is scoped to a single-vehicle routing game. This is a deliberate methodological choice — the single-tour setting isolates the temporal coalition-feasibility structure of $\mathcal{F}$ from fleet-assignment confounds, and the complement-based arguments rely on the single-vehicle insertion identity. A multi-vehicle *Online Vehicle Routing Game* — requiring $c(S)$ redefined with capacity and a fleet-assignment rule — is the subject of a companion manuscript in preparation.

Submission history disclosure

This manuscript was previously submitted to *Transportation Science* (TS-2026-0253) and was desk-rejected on scope grounds; the editor indicated the paper would need to include “more generalizable knowledge for the field of transportation science” before going out for review. We agree the cooperative-game-theoretic framing is closer to *EJOR*’s remit than to *Transportation Science*’s operational-decision focus, and accordingly resubmit to *EJOR* without retargeting the manuscript’s substance. The manuscript has not been published or submitted elsewhere.

Reproducibility

All code, random seeds, and raw CSV results are at <https://github.com/sage-Tae/online-tsg>. The repository ships version tags for each iteration in the revision history. `REPRODUCIBILITY.md` documents the environment, the full 175-instance main study (Table 4 of the manuscript), the 45-instance scale-up to $n = 50$, and the partition-pair-dual analysis script `code/scripts/intermediate_dual_analysis.py` used to extract the 9 partition-pair certificates of Supplementary §S9.

Declarations

- Declaration of interest: The authors declare no competing financial interests or personal relationships that could have influenced this work.
- The work reported has not been published elsewhere, and is not under consideration at any other journal.
- All authors have read and approved this submission.
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We thank the editorial office in advance for handling this manuscript and look forward to the reviewers’ feedback.

Sincerely,

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